CLIENT PRESENTATION

Customer Shopping Behavior Analysis

Spending patterns, customer segments and subscription behavior across 3,900 purchases

Python | PostgreSQL | Power BI

Dataset at a glance

3,900	$233K	$59.76	3.75
Purchases analysed	Total revenue	Average purchase	Average review rating

18 columns covering demographics, purchase details and shopping behavior. 37 missing review ratings were filled with the category median.

Source: customer shopping dataset, cleaned in Python and analysed in PostgreSQL

Revenue drivers

Revenue by gender



Male $157,890
Female $75,191

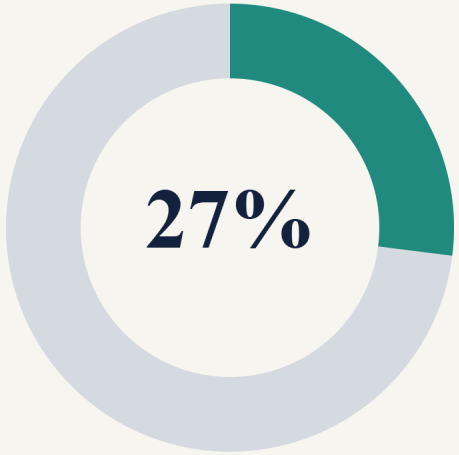
Revenue by age group



Male customers generate about two-thirds of revenue. Young Adults lead, but all four age groups sit within about 11% of each other.

Source: PostgreSQL queries on revenue by gender and age group

Subscription behavior



1,053 vs 2,847

Subscribers vs non-subscribers

$59.49 vs $59.87

Average spend per purchase, subscribers do not spend more

958 of 3,476

Repeat buyers (5+ previous purchases) who subscribe, about 28%

Source: PostgreSQL queries on subscription status and repeat buyers

Customer loyalty



3,116

Loyal, 80%

More than 10 previous purchases

701

Returning, 18%

2 to 10 previous purchases

83

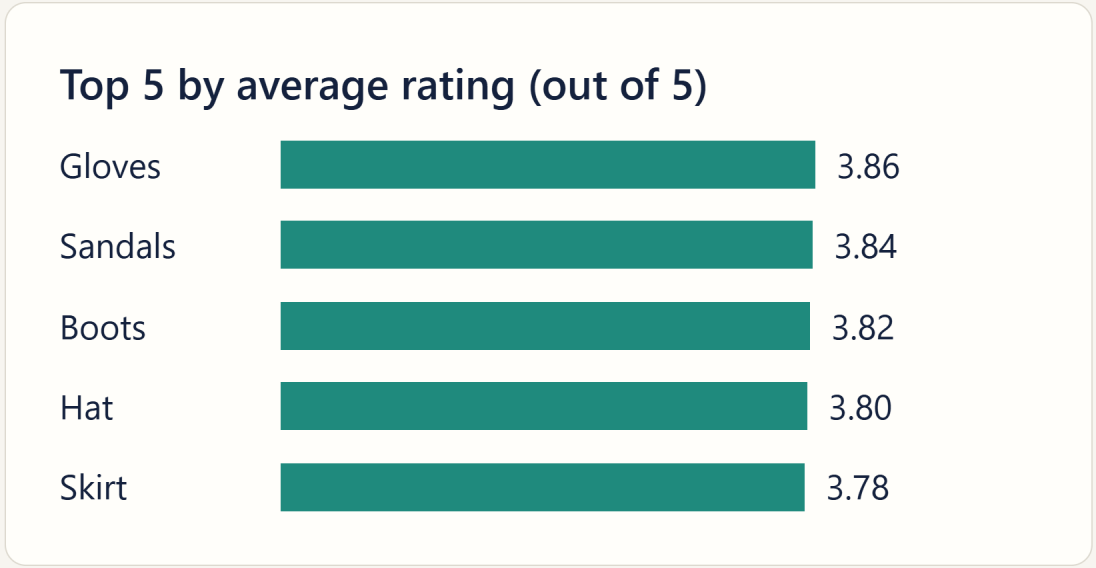
New, 2%

1 previous purchase

Express shipping orders average $60.48 against $58.46 for Standard, about 3% higher.

Source: PostgreSQL queries on customer segments and shipping type

Product performance





Hat appears on both lists: it is highly rated, and half of its purchases use a discount.

Source: PostgreSQL queries on product ratings and discount rates

Recommendations

1. **Boost subscriptions**
   Promote exclusive benefits, since subscribers are only 27% of customers.

2. **Customer loyalty programs**
   Reward repeat buyers to move them into the Loyal segment.

3. **Review discount policy**
   Balance sales boosts with margin control.

4. **Product positioning**
   Highlight top-rated and best-selling products in campaigns.

5. **Targeted marketing**
   Focus on high-revenue age groups and express-shipping users.